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Electrical power system converter control

Abstract

A electrical power system comprises: an electrical machine operable to output AC; a DC electrical network; a power electronics converter connected between the AC output of the electrical machine and the DC electrical network and including a plurality of transistors and associated diodes connected in parallel with the transistors; and a controller configured to control switching of the transistors of the converter so that, during normal operation of the electrical power system, the converter rectifies the AC output of the electrical machine to supply the DC electrical network with DC electrical power. The controller is further configured, responsive to a determination to the effect there is a fault in the DC electrical network, to control a voltage source, to inject a voltage to bias the diodes of the converter, and to control the switching of the transistors to control a level of current supplied to the faulted DC electrical network.

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Background/Summary

CROSS-REFERENCE TO RELATED APPLICATIONS

(1) This specification is based upon and claims the benefit of priority from United Kingdom Patent Application No. 2115015.6, filed on 20 Oct. 2021, the entire contents of which are incorporated herein by reference.

TECHNICAL FIELD

(2) This disclosure relates to electrical power systems, more particularly to the control of power electronics converters in electrical power systems. The electrical power systems may be of particular utility in transport applications including but not limited to aerospace.

BACKGROUND

(3) In aerospace, aircraft and their power and propulsion systems are becoming increasingly electric in their design. So-called 'more electric engines' (MEEs) and 'more electric aircraft' (MEAs) derive all or substantially all of their propulsive thrust from turbomachinery but make greater use of electrical power compared with conventional platforms. They may, for example, use electrical power to power auxiliary systems which have previously been powered mechanically or pneumatically, or may use spool-coupled electrical machines to transfer power to, from and between engine spools to provide improvements in engine operability and efficiency. In hybrid electric aircraft the propulsive thrust is derived from engines (e.g., gas turbine engines) and from other sources, typically batteries and/or fuel cells which supply electrical power to engine- or propulsor-coupled electrical machines.

(4) Some proposed more electric and hybrid electric platforms include DC electrical networks which receive electrical power from engine-driven electrical machines via AC to DC converters (i.e., rectifiers). The upper circuit of FIG. 4 illustrates a typical arrangement in which a three-phase electrical generator, which may be coupled with and driven by a spool of a gas turbine engine, is connected with a DC network via a two-level AC-DC converter. Each phase leg of the converter has two transistors (e.g., IGBTs), each of which is connected in anti-parallel with an associated diode. To rectify the AC output of the generator, for each phase leg current is commutated between the upper diode and the lower transistor, with each conducting for half of each AC cycle.

(5) The lower circuit of FIG. 4 illustrates the effect of a fault in the DC network which places a low impedance across the network terminals. In this case both the DC and AC sides of the converter

assume a low voltage condition, which in the limit may be zero Volts. In this condition there is no voltage to reverse bias the diodes, which means the AC-DC converter loses control and in effect reverts to being an uncontrolled diode rectifier. The level of current which will be supplied to the DC network in the faulted condition will therefore depend mainly on the voltage generated by and the impedance of the electrical machine.

(6) In some applications it is acceptable for the converter to operate as an uncontrolled diode rectifier when there is a DC network fault. In other applications, however, including aerospace applications, this mode of operation may be highly undesirable because the electrical machines may generate high voltages and/or may have low impedances, such that the high levels of fault current will be supplied to the fault site in the DC network.

(7) It would be desirable to be able to control the level of fault current supplied to the DC electrical network in the presence of a DC network fault.

SUMMARY

(8) The invention is directed to an electrical power system, a method of controlling an electrical power system and a controller for an electrical power system as set out in the appended claims.

(9) According to a first aspect, there is provided an electrical power system comprising: an electrical machine operable to output AC; a DC electrical network; a power electronics converter connected between the AC output of the electrical machine and the DC electrical network and comprising a plurality of transistors and associated diodes connected in parallel with the transistors; and a controller configured to control switching of the transistors of the converter so that, during normal operation of the electrical power system, the converter rectifies the AC output of the electrical machine to supply the DC electrical network with DC electrical power. The controller is further configured, responsive to a determination to the effect there is a fault in the DC electrical network, to control a voltage source to inject a voltage to bias the diodes of the converter, and to control the switching of the transistors to control a level of current supplied to the faulted DC electrical network.

(10) By responding to a DC network fault with the injection of a diode-biasing voltage the system retains the ability to commutate current between the diodes and transistors despite the collapse of the DC network voltage. This allows the converter to be switched into states in which the fault current can be reduced to zero, or at least to an average level below the full fault current level that would otherwise be conducted to the DC network. Not only does this protect loads in the DC network from a potentially very high fault current, it can provide an opportunity to perform protective measures, e.g., the opening of circuit breaking components (e.g., mechanical or semiconductor contactors or circuit breakers) in the DC network, whilst the network is subject to zero or low current. This may reduce problems associated with e.g., contactor arcing which can occur when components are actuated in high current conditions. Furthermore, the ability to control the DC side fault current may be particularly advantageous when a variable voltage and/or variable frequency generator (e.g., a permanent magnet generator) is employed as the fault current produced by the converter operating as an uncontrolled diode rectifier may vary significantly with generator shaft speed.

(11) The voltage source may comprise one or more of the following connectable with the power electronics converter and DC electrical network: a resistor; a capacitor; an inductor; and/or a semiconductor switching component. Said component(s) may be connected in series with (e.g., between) the DC output of the converter and the DC electrical network. Said component(s) may be connected in parallel with a controllable switch, whereby the component(s) may be selectively connected upon determination of a fault.

(12) Where the voltage source comprises a semiconductor switching component, controlling the voltage source to inject the voltage to bias the diodes may comprise operating the semiconductor switching component in a linear operating region to produce the biasing voltage. For example, a transistor may be operated for short periods of time in its linear region so as to develop a higher

than normal voltage to produce the diode commutating voltage.

(13) Where the voltage source comprises a capacitor, controlling the voltage source to inject the voltage to bias the diodes may comprise controlling the charge or discharge of the capacitor to produce the biasing voltage. For example, the voltage source may comprise a circuit (e.g., a bridge circuit such as a half-bridge or full-bridge) incorporating semiconductor switching components (e.g., transistors) and a capacitor. The controller may control the switching of the semiconductor switching components of the circuit so as to charge or discharge the capacitor to bias the diodes in response to the determination there is a fault in the DC network.

(14) The electrical power system may further comprise a soft-start circuit, and the soft-start circuit may comprise the voltage source. For example, the soft-start circuit may comprise a resistor and switch (e.g., a semiconductor switching device such as a transistor) connected in parallel. In response to the determination of the fault, the switch may be opened to connect the resistor to the converter and DC electrical network. In another example where the switch is a transistor, the transistor may be operated for short periods of time in its linear region so as to develop a higher-than-normal voltage to produce the diode commuting voltage. The use of a soft-start circuit to provide the voltage source function may be particularly convenient because the electrical power system may already include a soft-start circuit. Thus, no additional components or equipment need be provided.

(15) Controlling the switching of the transistors to control the level of current supplied from the electrical machine to the faulted DC electrical network may comprise: switching the transistors of the converter to a crow-bar configuration in which current from the electrical machine does not flow to the DC network. Zero fault current is supplied to the DC network in the crow-bar configuration and the current is instead contained within the converter (e.g., within one half of the bridge of a bridge-configured converter).

(16) Controlling the switching of the transistors to control the level of current supplied from the electrical machine to the faulted DC electrical network may comprise: repeatedly switching the transistors between the crow-bar configuration and a rectifier configuration in which current from the electrical machine does flow to the DC network. In this case the time spent in the crow-bar configuration reduces the time-averaged current supplied to the DC network.

(17) Controlling the switching of the transistors to control the level of current supplied from the electrical machine to the faulted DC electrical network may comprise: modulating a pulse width modulation (PWM) control parameter to control the level of current supplied from the electrical machine to the faulted DC electrical network. For example, a frequency and/or pulse width (i.e., duty cycle) of a PWM signal used to control the switching of the transistors may be adjusted to control (e.g., reduce) the level of current supplied to the DC network.

(18) The electrical power system may further comprise a filter for smoothing a waveform of the current supplied to the faulted DC electrical network. The filter may be a low-pass filter.

(19) The diodes of the converter may be separate from the transistors, with the transistors being of any suitable known type (e.g., IGBTs, MOSFETs). In other examples the diodes may be body diodes of MOSFETs. Those skilled in the art will understand the term “body diode” to be the diode character inherent to a MOSFET, which can effectively be considered to be a diode connected in parallel with the bi-directional switchable conduction path between the source and drain of the MOSFET.

(20) The electrical machine may be of any suitable type. The electrical machine may for example be a permanent magnet machine, a wound field machine, or a switched reluctance machine.

(21) The electrical machine may have a plurality of phases; and the power electronics converter may comprises a phase leg for each one of the plurality of phases of the electrical machine. Each phase leg may comprise: a first transistor and associated parallel diode connected between a phase connection of the electrical machine and a first DC output of the converter; and a second transistor and associated parallel diode connected between the phase connection of the electrical machine and

a second DC output of the converter. The first and second DC outputs of the converter may be connected with the DC electrical network. Any number of phases greater than or equal to two may be used. In one specific example the number of phases is three, and in another specific example the number of phases is four.

(22) According to a second aspect, there is provided a method of controlling an electrical power system comprising an electrical machine, a DC electrical network and a power electronics converter connected between the electrical machine and the DC electrical network. The method comprises: generating AC using the electrical machine and supplying the AC to an input of the converter; controlling switching of transistors of the converter to rectify the AC input to a DC output and supply the DC output to the DC electrical network; determining a condition to the effect that there is a fault in the DC electrical network; responsive to the determination, controlling a voltage source to inject a voltage to bias diodes of the converter; and after controlling the voltage source to inject a voltage to bias diodes of the converter, controlling switching of the transistors of the converter to control a level of current supplied to the DC electrical network.

(23) Controlling switching of the transistors of the converter to control the level of current supplied to the DC electrical network may comprise: switching the transistors of the converter to a crow-bar configuration in which current from the electrical machine does not flow to the DC network.

(24) Controlling switching of the transistors of the converter to control the level of current supplied to the DC electrical network may comprise: repeatedly switching the transistors between the crow-bar configuration and a rectifier configuration in which current from the electrical machine does flow to the DC network.

(25) Controlling switching of the transistors of the converter to control the level of current supplied to the DC electrical network may comprise: modulating a pulse width modulation control parameter to control the level of current supplied from the electrical machine to the faulted DC electrical network.

(26) The method may further comprise, after controlling switching of the transistors of the converter to control the level of current supplied to the DC electrical network: opening one more circuit breakers or contactors in the DC electrical network to isolate the fault in the DC network. In this way the fault and/or other DC network components can be isolated. By opening the contactor after controlling the level of current, problems such as contactor arcing can be reduced. Such problems may be particularly reduced if the opening is performed at zero current, e.g., when the converter is in the crow-bar configuration.

(27) According to a third aspect, there is provided a controller for an electrical power system configured to: during normal operation of the electrical power system, control switching of a plurality of transistors of a power electronics converter to rectify AC received by the converter from an electrical machine; and responsive to a determination to the effect there is a fault in a DC electrical network connected to a DC side of the converter, control a voltage source to inject a voltage to bias diodes of the converter, and control the switching of the transistors to control a level of current supplied to the faulted DC electrical network.

(28) The controller may be implemented as a single controller or multiple separate (e.g., distributed) controllers. Thus the controller may be or may form part of a control system. The controller may be implemented in software, hardware or a combination of the two. The controller may be or may be a functional module of an Engine Electronic Controller (EEC) or a Full Authority Digital Engine Controller (FADEC).

(29) According to a fourth aspect, there is provided an aircraft power and propulsion system comprising: a gas turbine engine; and an electrical power system according to the first aspect. The electrical machine of the electrical power system is mechanically coupled with a spool of the gas turbine engine. The power electronics converter may be a unidirectional AC to DC converter (i.e., a rectifier) or a bidirectional AC-DC converter capable of operating as either a rectifier or an inverter depending on an operating mode of the electrical machine.

(30) According to a fifth aspect, there is provided an aircraft comprising the power and propulsion system of the fourth aspect. The aircraft may be a solely gas-turbine-powered aircraft (e.g., a more electric aircraft) or a hybrid electric aircraft.

(31) The skilled person will appreciate that except where mutually exclusive, a feature described in relation to any one of the above aspects may be applied mutatis mutandis to any other aspect. Furthermore, except where mutually exclusive, any feature described herein may be applied to any aspect and/or combined with any other feature described herein.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

- (1) Embodiments will now be described by way of example only with reference to the accompanying drawings, which are purely schematic and not to scale, and in which:
- (2) FIG. 1 shows a general arrangement of a turbofan engine for an aircraft;
- (3) FIG. 2A is a schematic illustration of a hybrid electric aircraft propulsion system;
- (4) FIG. 2B illustrates an electrically powered propulsor such as may be used in a hybrid electric propulsion system;
- (5) FIG. 3A illustrates a portion of an exemplary electrical power system in which an AC-DC converter connects a 3-phase electrical generator with a DC network;
- (6) FIG. 3B illustrates how current is switched between the upper and lower branches of a phase leg of the converter to rectify AC to DC;
- (7) FIG. 4 illustrates the effect of a DC network fault on an AC-DC converter;
- (8) FIG. 5A illustrates the electrical power system of FIG. 3A incorporating a soft-start circuit;
- (9) FIG. 5B illustrates how the soft-start circuit can inject a diode biasing voltage in the presence of a DC network fault;
- (10) FIG. 5C illustrates the switching of the converter into a crow-bar configuration following a DC network fault;
- (11) FIGS. 6A-C illustrate an arrangement in which a diode biasing voltage is injected by discharging a capacitor of a full-bridge circuit;
- (12) FIG. 7 illustrates the waveform of the fault current which will be supplied to the DC network following a fault in the uncontrolled diode rectifier mode;
- (13) FIG. 8 illustrates how the waveform of the fault current can be modified using the crow-bar configuration, modulation of PWM control parameters and a filter; and
- (14) FIG. 9 is a flow chart illustrating a method of controlling an electrical power system.

DETAILED DESCRIPTION

(15) FIG. 1

(16) A general arrangement of an engine **101** for an aircraft is shown in FIG. 1. The engine **101** is of turbofan configuration, and thus comprises a ducted fan **102** that receives intake air A and generates two pressurised airflows: a bypass flow B which passes axially through a bypass duct **103** and a core flow C which enters a core gas turbine.

(17) The core gas turbine comprises, in axial flow series, a low-pressure compressor **104**, a high-pressure compressor **105**, a combustor **106**, a high-pressure turbine **107**, and a low-pressure turbine **108**.

(18) In operation, the core flow C is compressed by the low-pressure compressor **104** and is then directed into the high-pressure compressor **105** where further compression takes place. The compressed air exhausted from the high-pressure compressor **105** is directed into the combustor **106** where it is mixed with fuel and the mixture is combusted. The resultant hot combustion products then expand through, and thereby drive, the high-pressure turbine **107** and in turn the low-pressure turbine **108** before being exhausted to provide a small proportion of the overall thrust.

(19) The high-pressure turbine **107** drives the high-pressure compressor **105** via an interconnecting shaft. The low-pressure turbine **108** drives the low-pressure compressor **104** via another interconnecting shaft. Together, the high-pressure compressor **105**, high-pressure turbine **107**, and associated interconnecting shaft form part of a high-pressure spool of the engine **101**. Similarly, the low-pressure compressor **104**, low-pressure turbine **108**, and associated interconnecting shaft form part of a low-pressure spool of the engine **101**. Such nomenclature will be familiar to those skilled in the art. Those skilled in the art will also appreciate that whilst the illustrated engine has two spools, other gas turbine engines have a different number of spools, e.g., three spools.

(20) The fan **102** is driven by the low-pressure turbine **108** via a reduction gearbox in the form of a planetary-configuration epicyclic gearbox **109**. Thus in this configuration, the low-pressure turbine **108** is connected with a sun gear of the gearbox **109**. The sun gear is meshed with a plurality of planet gears located in a rotating carrier, which planet gears are in turn meshed with a static ring gear. The rotating carrier drives the fan **102** via a fan shaft **110**. It will be appreciated that in alternative embodiments a star-configuration epicyclic gearbox (in which the planet carrier is static and the ring gear rotates and provides the output) may be used instead, and indeed that the gearbox **109** may be omitted entirely so that the fan **102** is driven directly by the low-pressure turbine **108**.

(21) It is increasingly desirable to facilitate a greater degree of electrical functionality on the airframe and on the engine. To this end, the engine **101** of FIG. **1** comprises one or more rotary electrical machines, generally capable of operating both as a motor and as a generator. The number and arrangement of the rotary electrical machines will depend to some extent on the desired functionality. Some embodiments of the engine **101** include a single rotary electrical machine **111** driven by the high-pressure spool, for example by a core-mounted accessory drive **112** of conventional configuration. Such a configuration facilitates the generation of electrical power for the engine and the aircraft and the driving of the high-pressure spool to facilitate starting of the engine in place of an air turbine starter. Other embodiments, including the one shown in FIG. **1**, comprise both a first rotary electrical machine **111** coupled with the high-pressure spool and a second rotary electrical machine **113** coupled with the low pressure spool. In addition to generating electrical power and the starting the engine **101**, having both first and second rotary machines **111**, **113**, connected by power electronics, can facilitate the transfer of mechanical power between the high and lower pressure spools to improve operability, fuel consumption etc.

(22) As mentioned above, in FIG. **1** the first rotary electrical machine **111** is driven by the high-pressure spool by a core-mounted accessory drive **112** of conventional configuration. In alternative embodiments, the first electrical machine **111** may be mounted coaxially with the turbomachinery in the engine **101**. For example, the first electrical machine **111** may be mounted axially in line with the duct between the low- and high-pressure compressors **104** and **105**. In FIG. **1**, the second electrical machine **113** is mounted in the tail cone **114** of the engine **101** coaxially with the turbomachinery and is coupled to the low-pressure turbine **108**. In alternative embodiments, the second rotary electrical machine **113** may be located axially in line with low-pressure compressor **104**, which may adopt a bladed disc or bladed drum configuration to provide space for the second rotary electrical machine **113**. It will of course be appreciated by those skilled in the art that any other suitable location for the first and (if present) second electrical machines may be adopted.

(23) The first and second electrical machines **111**, **113** are connected with power electronics. Extraction of power from or application of power to the electrical machines is performed by a power electronics module (PEM) **115**. In the present embodiment, the PEM **115** is mounted on the fan case **116** of the engine **101**, but it will be appreciated that it may be mounted elsewhere such as on the core of the gas turbine, or in the vehicle to which the engine **101** is attached, for example.

(24) Control of the PEM **115** and of the first and second electrical machines **111** and **113** is in the present example performed by an engine electronic controller (EEC) **117**. In the present embodiment the EEC **117** is a full-authority digital engine controller (FADEC), the configuration of which will be known and understood by those skilled in the art. It therefore controls all aspects of

the engine **101**, i.e., both of the core gas turbine and the first and second electrical machines **111** and **113**. In this way, the EEC **117** may holistically respond to both thrust demand and electrical power demand.

(25) The one or more rotary electrical machines **111**, **113** and the power electronics **115** may be configured to output to or receive electric power from one, two or more dc busses. The dc busses allow for the distribution of electrical power to other engine electrical loads and to electrical loads on the airframe.

(26) Those skilled in the art will appreciate that the gas turbine engine **101** described above may be regarded as a 'more electric' gas turbine engine because of the increased role of the electrical machines **111**, **113** compared with those of conventional gas turbines.

(27) FIGS. **2A** and **2B**

(28) FIG. **2A** illustrates an exemplary propulsion system **200** of a hybrid electric aircraft. The propulsion system **200** includes a generator set **202** comprising an engine **201** and electrical generator **211**, and a battery pack **203**. Both the generator set **202** and the battery pack **203** are used as energy sources to power a motor-driven propulsor **204**, an example of which is shown in FIG. **2B**.

(29) The illustrated propulsion system **200** further comprises an AC/DC converter **205**, a dc distribution bus **210**, a DC/AC converter **206** and a DC/DC converter **207**. It will be appreciated that whilst one generator set **202** and one propulsor **204** are illustrated in this example, a propulsion system **200** may include more than one generator set **202** and/or one or more propulsor **204**.

(30) A shaft or spool of the engine **201** is coupled to and drives the rotation of a shaft of the generator **211** which thereby produces alternating current. The AC/DC converter **205**, which faces the generator **211**, converts the alternating current into direct current which is fed to various electrical systems and loads via the dc distribution bus **210**. These electrical systems include non-propulsive loads (not shown in FIG. **2A**) and the motor **213** which drives the propulsor **204** via the DC/AC converter **206**.

(31) The battery pack **203**, which may be made up of a number of battery modules connected in series and/or parallel, is connected to the dc distribution bus **210** via the DC/DC converter **207**. The DC/DC converter **207** converts between a voltage of the battery pack **203** and a voltage of the dc distribution bus **210**. In this way, the battery pack **203** can replace or supplement the power provided by the generator set **202** (by discharging and thereby feeding the DC distribution bus **210**) or can be charged using the power provided by the generator set **202** (by being fed by the dc distribution bus **210**).

(32) Referring to FIG. **2B**, in this example the propulsor **204** takes the form of a ducted fan. The fan **216** is enclosed within a fan duct **219** defined within a nacelle **221**, and is mounted to a core nacelle **215**. The fan **216** is driven by the electrical machine **213** via a drive shaft **214**, both of which may also be thought of as components of the propulsor **204**. In this embodiment a gearbox **220** is provided between the electrical machine **213** and the drive shaft **214**.

(33) The electrical machine **213** is supplied with electric power from a power source, for example the generator set **202** and/or the battery **203** via the dc bus **210**. The electrical machine **213** of the propulsor, and indeed the electrical machine **211** of the generator set **202**, may be of any suitable type, for example of the permanent magnet synchronous type.

(34) Those skilled in the art will recognise the propulsion system **200** of FIGS. **2A-B** to be of the series hybrid type. Other hybrid electric propulsion systems are of the parallel type, while still others are of the turboelectric type or have features of more than one type. The configuration of the more electric engine **101** of FIG. **1** may be considered similar to a parallel hybrid system, with the main distinction being the roles of the electrical machines. For example, the electrical machines of a more electric engine are generally only used in motor mode to start the engine and to improve engine operability, whereas the electric machines of a parallel hybrid propulsion system are used to motor the spools to meaningfully add to the amount of propulsive thrust produced by the

turbomachinery.

(35) Those skilled in the art will also appreciate that the hybrid architecture illustrated in FIG. 2A is only one example, and other architectures are known and will occur to those skilled in the art.

(36) FIGS. 3A-B

(37) FIG. 3A illustrates a portion of an electrical power system **300** such as may be used in the aircraft and engine systems described above with reference to FIG. 1 and FIGS. 2A-B. The electrical power system **300** includes an electrical machine **310** operable as a generator to generate AC, an AC-DC power electronics converter **320** operable to rectify the AC received from the electrical machine **310** and output DC, a DC electrical network **330** which receives the DC power output by the converter **320**, and a controller **340** which exercises control over the electrical power system **300** including the converter **320**.

(38) In this example the electrical machine **310** is a three-phase machine which outputs its three phases through three phase connections **311a**, **311b**, **311c**. It will however be understood that other numbers of phases (e.g., four) and other connection arrangements could be used. The electrical machine **310** can be of any type and configuration suitable for the application requirements. In one specific embodiment it is a permanent magnet synchronous machine (PMSM) of radial flux configuration, but other types of machine including wound field and switched reluctance, and other configurations such as axial flux and transverse flux, are also contemplated. The electrical machine **310** may be specifically configured as a generator, or it may be operable in both motor and generator modes (e.g., where the electrical machine **310** is a 'starter-generator' of a gas turbine engine).

(39) The illustrated power electronics converter **320** is a two-level, three-phase full-bridge rectifier with insulated gate bipolar transistors (IGBTs). It includes three phase legs **321a**, **321b**, **321c**, each of which is connected to a corresponding one of the phase connections **311a**, **311b**, **311c** from which it receives AC. Each phase leg **321a**, **321b**, **321c** includes two branches: a first (upper) branch which includes a transistor **322** and associated diode **323** connected in anti-parallel with the transistor **322** and a second (lower) branch which includes a transistor **324** and associated diode **325** connected in anti-parallel with the transistor **324**. The upper branch of each phase leg **321a**, **321b**, **321c** is connected with a first DC output **326**. The lower branch of each phase leg **321a**, **321b**, **321c** is connected with a second DC output **327**. The first and second DC outputs **326**, **327** connect with the DC electrical network **330**, whereby the DC electrical network **330** is supplied with DC electrical power. A DC link capacitor **328** is also shown, the function of which will be familiar to those skilled in the art of power electronics.

(40) Whilst a two-level, three-phase rectifier **320** with IGBTs is illustrated, this is not intended to limit the invention. Different numbers of phases and different converter topologies may be used. The converter **320** may also utilise another type of transistor, for example MOSFETs (e.g., wide band gap semiconductor MOSFETs such as Silicon Carbide (SiC) MOSFETs), especially if the converter is bidirectional to allow for both rectification and inversion of current. Where MOSFETs are used, the anti-parallel diodes **323**, **325** may be retained or alternatively omitted in place of the MOSFET's inherent body diode, as will be understood by those skilled in the art. However generally speaking it is preferable to retain the anti-parallel diode, even if MOSFETs are used, as the additional diode may permit e.g., faster switching, higher current rating and lower on-state voltage drop.

(41) During normal operation of the converter **320** as a rectifier, the controller **340** uses pulse width modulation to control the switching of the transistors to commutate current between the upper and lower branches of the phase legs to affect suitable rectification of the AC to DC. In this example, when current flows through the upper branch of a phase leg to the first output **326** it flows through the diode **322**, whereas when current flows through the lower branch of the phase leg to the second output **327** it flows through the transistor **324**.

(42) FIG. 3B shows, for one phase leg, the normal process of commutating from an upper diode

323 to a lower transistor **324**. Initially current is flowing through the upper leg through the upper diode **323**. By turning on the lower transistor **324** (as in the right-hand circuit), the DC side voltage is imposed across the conducting diode **323** in a reverse direction, which acts to turn off the diode **323** allowing conduction through the lower transistor. Thus, it can be seen that the DC side voltage is instrumental in commutating the diode **323** to its non-conducting state to permit commutation of the current to the lower branch.

(43) FIG. 4

(44) FIG. 4 illustrates the effect of a DC network fault (e.g., a fault in a load connected in the DC network which places a low impedance across the DC network terminals) on the operation of the electrical power system **300**.

(45) Following a fault in the DC network **330** the DC-side voltage collapses, possibly to zero Volts though in general the fault may have some resistance in which case the voltage may not collapse all the way to zero. The significance of this is that there is no voltage (or more generally only a small voltage) available to reverse bias the conducting diode (e.g., diode **323** in FIG. 3B) to turn it off. If the voltage is not large enough (and in general this cannot be guaranteed), the converter loses the ability to commute current between the upper and lower branches of the phase legs **321a**, **321b**, **321c**.

(46) Another effect of the DC network fault is that the fault current supplied to the DC network **330** will be controlled almost exclusively by the voltage generated by the electrical machine **310** and the machine's impedance. The fault current supplied to the DC network **330** via the converter **320** will therefore typically be very large. For example, even for a 3-phase generator deliberately designed to have a high reactance to limit the output fault current of each phase to 500 A.sub.RMS, the DC fault current contributed by 3-phases acting together would be approximately 707 A and would also contain a significant ripple component.

(47) In view of the high fault current, it is common to switch off the active transistors in the event of a DC network fault, to protect them from over-current stress. This leaves the converter configuration illustrated in the lower circuit of FIG. 4: the converter **320** will operate as an uncontrolled diode rectifier. Briefly referring to FIG. 7, this illustrates the 3-phase generator current output by the generator **310** and also the waveform of the fault current supplied to the DC network **330** in the uncontrolled rectifier mode of operation.

(48) In some systems the uncontrolled diode rectifier mode of operation is acceptable because the converter **320** is at least supplying current to the network **330**. In other systems, however, in particular safety-critical aerospace systems, it would be highly desirable to have additional control over the level of current supplied to the DC network **330** following the fault.

(49) FIGS. 5A-C

(50) In accordance with an embodiment of the invention, FIGS. 5A and 5B illustrate the use of a soft-start circuit **350** to inject a voltage to reverse bias the diodes of the converter **320** to allow control of the commutation of current despite a DC network fault. FIG. 5C illustrates the commutation of the converter **320** into a crow-bar configuration to reduce the fault current supplied to the DC network **330** following the fault.

(51) In more detail, FIG. 5A illustrates the previously described electrical power system **300** but with a soft-start circuit **350** also shown connected on the DC-side of the converter **320**. The soft-start circuit, which is well-known and widely used in electrical power systems, comprises a resistor **351** with a bypass switch **352** connected across its terminals. The bypass switch **352** can take any suitable form. In some embodiments the switch **352** comprises one or more semiconductor switching components, for example transistors or silicon controlled rectifiers (SCRs).

(52) When the converter **320** is first energised, during which time the DC filter capacitor **328** is initially charged, the DC network **330** will tend to supply a large and damaging inrush current. The resistor **351** of the soft-start circuit **350** acts to limit this current to prevent damage. Once stable operation is reached (e.g., when the DC filter capacitor **328** is charged), the resistor **351** is shorted

out by closing the bypass switch **352**. This prevents the resistor **351** negatively affecting operation of the converter **320**. (If not shorted out the resistor **351** would e.g., cause the DC output voltage to adversely change with load current and also create additional energy losses).

(53) Ordinarily this is the extent of the operation of the soft-start circuit **350**. However, in accordance with some embodiments described herein, the resistor **351** is reintroduced to the circuit (by controlling the opening the switch **352**) in the event of a fault in the DC network **330**. This is illustrated in FIG. 5B. Reintroducing the resistor **351** and its resistance to the fault current path decouples the DC filter capacitor **328** from the fault, directing a proportion of the fault current to flow into the capacitor **328** thus allowing it to develop a direct voltage, shown in FIG. 5B, capable of suitably biasing the converter diodes.

(54) With the diode biasing DC-side voltage established, the converter **320** can commutate current between the branches of the phase legs **321a-c** and thus remove current from the DC fault. This is illustrated in FIG. 5C. As shown in upper circuit of FIG. 5C, immediately following the fault the converter **320** enters the diode rectifier mode. In response, and as is shown the lower circuit, the controller **340** opens the bypass switch **352** to introduce the resistor **351** to the circuit to reverse bias the converter diodes. The controller **340** then controls the switching of the transistors to commutate the current so as to enter the crow-bar configuration. In the crow-bar configuration the current from the generator **310** cannot reach the DC network **320** and is instead contained within the generator **310** and converter **320**.

(55) It will be appreciated that FIG. 5C illustrates a crow-bar configuration in which current is conducted through the lower set of transistor-diode pairs (i.e., those labelled 4, 6 and 2) such that current is contained in the lower half of the converter bridge. Alternatively, a crow-bar configuration in which current is conducted through the upper set of transistor-diode pairs (i.e., those labelled 1, 3 and 5) could be used, or indeed the controller may switch the converter **320** between these two configurations.

(56) It is generally undesirable for the resistor **351** of the soft-start circuit **345** to be connected within the DC fault path for prolonged periods, as the resistor **351** will dissipate a large amount of heat and may be damaged unless adequate heat sinking is provided. However, the injection of the resistor **351** for short periods to assist in the commutation of the converter **320** should generally be acceptable. In many cases the resistor **351** need only be switched into the circuit for long enough to affect a new system operating state, for example by opening or closing one or more circuit breakers (e.g., DC contactors) in the DC network to isolate the fault. Operating the circuit breakers at low or zero fault current is desirable as it may, for example, alleviate problems with contactor arcing.

(57) In another embodiment of the invention, where the bypass switch **352** of the soft-start circuit **350** takes the form of a semiconductor switching component (e.g., a transistor) the injection of the voltage to bias the diodes of the converter **320** may be performed without the use of the resistor **351**. Specifically, operating the semiconductor switching component **352** for short periods of time in its linear operating region, with a current level within its safe thermal limits, will develop a higher voltage to produce the diode commuting voltage.

(58) Whilst the embodiments described above envisage the use of components of a soft-start circuit **350** to inject the biasing voltage, it should be appreciated that the resistor **351** and/or semiconductor switching component **352** used to inject the voltage may instead form part of another arrangement. For example, one or both components **351**, **352** may be part of an auxiliary circuit dedicated to providing a biasing voltage following a DC network fault. Using the soft-start circuit **350** may be considered particularly convenient because it allows the biasing function to be achieved without the addition of system components, since many systems already include a soft-start circuit **350**. However, the use of another (e.g., dedicated) circuit may allow for the selection of components with parameters specifically selected for the voltage injection function (e.g., a particular resistance value which balances the need to provide a high enough voltage but which does not generate excessive heat may be selected).

(59) FIGS. 6A-C

(60) FIGS. 6A to 6C illustrate another embodiment of the invention by which a diode biasing voltage may be injected in response to a DC network fault. It is noted that FIGS. 6A-C are a simplified view of the system 300 and the electrical machine 310 is not shown.

(61) In this embodiment an auxiliary voltage source 360 is connected between the DC side of the converter 320 and the DC network 330. The auxiliary voltage source 360 comprises a capacitor 361 connected across the terminals of a low voltage bridge 362 (a full-bridge or half-bridge structure may be used, for example) which includes a number of transistors. The transistors can be switched to either isolate or connect the capacitor 361 to the DC side of the converter 320.

(62) FIG. 6B illustrates the normal (i.e., un-faulted) operation of the system 300. In this case the transistors of the low voltage bridge 362 are switched so as to apply a voltage to charge the capacitor 361 and to also provide a bypass path so that the capacitor is not connected with the DC network 330 and converter 320.

(63) In response to a fault in the DC network 330, the controller 340 controls the switching of the transistors of the bridge 362 so as to connect the capacitor 361 between the DC side of the converter 320 and the DC network 330. This is shown in FIG. 6C. The inserted capacitor 361, which in this example is charged by the fault current to develop an increased voltage, provides the desired biasing voltage to the converter diodes.

(64) It should be appreciated that voltage sources 350, 360 described with reference to FIGS. 5A-C and 6A-C are only examples and that other ways of providing the desired voltage will occur to those skilled in the art. Generally speaking, any suitable circuit arrangement comprising one or more resistors, capacitors, inductors or semiconductor components capable of being switched into and out of DC path may be used.

(65) FIG. 7

(66) The left-hand graph of FIG. 7 illustrates the 3-phase AC-side current produced by a 3-phase electrical generator 310. The three phases are sinusoidal and offset from each other by 120 degrees. One 60 degree interval of one of the three phases (phase A) is highlighted within the dashed box.

(67) The right-hand graph of FIG. 7 illustrates the corresponding fault current delivered to the DC network 330 when the converter 320 operates as an uncontrolled diode rectifier, i.e., without utilising the invention described herein. The same 60 degree interval is highlighted in the right-hand graph.

(68) FIG. 8

(69) FIG. 8 illustrates ways in which the current which is supplied to the faulted DC electrical network may be controlled. By way of reference the top-left waveform of FIG. 8 (labelled "No PWM") is the fault current waveform in the absence of any control (i.e., it corresponds to the right-hand graph of FIG. 7 in which the invention is not utilised).

(70) According to one embodiment, following the injection of the diode biasing voltage, the transistors of the converter 320 are switched into the crow-bar configuration shown in FIG. 5C. This results in zero current being supplied to the DC network. Supplying zero current to the faulted DC network may be desirable in some instances, for example where circuit breaking components such as DC contactors are to be opened or closed to isolate faults.

(71) In other embodiments, where it is instead preferable to supply some (i.e., not zero) current to the DC network, the controller 340 may repeatedly switch the converter transistors between the crow-bar configuration and a rectifier configuration in which the fault current is supplied to the DC network. In this way the time-averaged current supplied to the DC network will be less than full fault current level because the time spent in the crow-bar configuration will pull the average current down.

(72) Where the latter approach is taken, the average current and the current waveform will depend on the modulation of the PWM signal used to switch the transistors. The greater the pulse width (equivalently, the greater the duty cycle), the higher the average current will be. The greater the

frequency, the more chopped the waveform will be. The latter is illustrated in the second and third left-hand waveforms of FIG. 8 (labelled “low frequency PWM” and “high frequency PWM”). Thus, the PWM control parameters may be selected and varied according the desired current level and waveform.

(73) Optionally, the current output by the converter **320** can be passed through an existing or dedicated filter **370** (e.g., a low-pass DC filter) to smooth the current waveform supplied to the DC network **330**. In the presence of a fault the DC filter may be tuned according to the downstream requirements of the DC network. The unfiltered waveforms are illustrated on the left-hand side of FIG. 8, under the heading “Unfiltered”. An exemplary filtered waveform is shown on the right-hand side under the heading “Filtered”.

(74) FIG. 9

(75) FIG. 9 is a flow diagram illustrating a method **10** of controlling an electrical power system **300**. The method **10** may be performed by a control system which is or includes the controller **340**. The controller **340** may, for example, be an EEC **117** or one or more functional modules of an EEC **117**.

(76) At step **11** the electrical power system **300** is operating in a normal condition. The rotor of the electrical machine **310** is being driven to rotate by a shaft, for example a spool of a gas turbine engine, and the machine **310** is generating AC electrical power. The generated AC electrical power is output from the electrical machine to an AC to DC power electronics converter **320**. Generally speaking the electrical machine **310** may be a variable speed machine driven to rotate at or at a speed related to the speed of the engine drive shaft, such that the frequency and voltage of the output AC varies.

(77) At step **12**, with the electrical power system **300** still operating in the normal condition, the AC to DC converter **320** rectifies the AC to DC and outputs the DC to supply the DC electrical network **330** with DC electrical power. In order to rectify the AC to DC, the controller **340** controls the switching of the transistors of the phase legs of the converter to commutate current and affect rectification. For example, the controller **340** may supply PWM signals to the transistors in an appropriate order and with appropriate switching frequency and duty cycle to supply the desired level of DC to the network **330**.

(78) At step **13**, the controller **340** determines a condition to the effect that there is a fault in the DC electrical network **330**. The determination of a fault may occur in any number of different ways, and the precise details are beyond the scope of the present disclosure. In one example, voltage and current levels are measured at the loads and at a various other positions about the network **330**, and faults are recognised by the controller **340** based on changes in the measured current and/or voltage levels. Alternatively, fault detection may take place locally (e.g., at a controller local to a faulted load) and a fault report may then be communicated to the controller **340**. Various other ways of determining faults are known and will occur to those skilled in the art.

(79) At step **14**, responsive to the determination in step **13**, the controller **340** controls a voltage source **350**, **360** to inject a voltage to bias the diodes of the converter **320**. In particular, the fault in the DC network **330** will have caused the DC-side voltage to collapse, likely to a level below that required to reverse bias the converter diodes to allow commutation of the current between the branches of the converter phase legs. The controller **340** therefore controls a voltage source **350**, **360** to inject a voltage sufficient to appropriately bias the diodes.

(80) The voltage source **350**, **360** can take any suitable form which allows an appropriate voltage to be controllably injected from the DC side of the converter **320**. In some embodiments the voltage source takes the form of a dedicated auxiliary circuit **360** whose specific purpose is to selectively inject voltage from the DC side of the converter **320** in case of a fault. In other examples an existing circuit, for example a soft-start circuit **350**, may be leveraged to inject the voltage so that further components are not required to support the function. Various examples of voltage sources have been described above with reference to FIGS. 5A-5C and 6A-6C.

(81) At step **15**, after controlling the voltage source **350**, **360** to inject a voltage to reverse bias the diodes in step **14**, the controller **340** controls switching of the transistors of the converter **320** to control a level of current supplied to the DC electrical network. In particular, with the ability to commutate the current restored, the controller **340** can control the converter **320** to reduce the level of current supplied to the DC network **330** to a level below the full fault current which would otherwise be supplied.

(82) In some embodiments the controller **340** may control the converter **320** to reduce the current level to zero, by switching to and remaining in the crow-bar configuration. In other embodiments the controller **340** may control the converter to reduce the current level to a level between zero and the full fault current, by switching between the crow-bar configuration and the rectifier configuration. In this case the current level and waveform can be controlled by controlling PWM control parameters, for example the frequency and duty cycle of the PWM signals. The DC output may also be passed through a filter to smooth the waveform to better approximate a true DC signal.

(83) It should be appreciated that if bidirectional MOSFETs are used in place of IGBTs, the diode rectifier mode illustrated in FIG. 5C may be replaced by a so-called 'synchronous rectification mode' which makes use of the bi-directional current carrying capability of the MOSFETs. In the synchronous rectification mode, the diodes (whether separate diodes or the inherent body diodes of the MOSFETs) do not carry current. Instead, the bi-directional MOSFETs carry current in their reverse directions in place of the diodes. For instance, referring to FIG. 3A, current of phase C flowing through the upper branch of the third phase leg **321c** to the first DC output **326** would flow through the transistor **322** in its reverse direction, and would not flow through the diode **323** as it would in the diode rectifier mode. Current of phase C flowing through the lower branch of the third phase leg **321c** to the second DC output **327** would flow through the transistor **324** as in the diode rectifier mode. Those skilled in the art will appreciate the synchronous rectification mode may also be used prior to the fault, during step **12**.

(84) Optionally, at step **16**, after controlling the current level at step **15**, the controller **340** controls the state of one or more circuit breaking components (e.g., contactors) in the DC network **330**. For example, the controller **340** may open a DC contactor to isolate a fault from the remainder of the DC network **330**. If it is possible to completely isolate the fault, it may be possible to return to the normal operating condition of the system, with full and normal control of the AC to DC converter **320**. In other cases the system **300** may remain in the condition of step **15** in which the current level is controlled but a fault still remains. In some instances, having safely controlled the current level in the DC network and taken appropriate action, the generator **310** may be disconnected from the drive shaft or may be isolated from the remainder of the electrical power system **300**.

(85) Various examples have been described, each of which feature various combinations of features. It will be appreciated by those skilled in the art that, except where clearly mutually exclusive, any of the features may be employed separately or in combination with any other features and the invention extends to and includes all combinations and sub-combinations of one or more features described herein.

(86) It should also be noted that whilst the embodiments have been described with reference to an aircraft, and to turbofan engines, it will be understood that the principles of the described electrical systems may be applied to other installations, for example to aircraft with turboprop engines, aircraft with purely electric power and propulsion systems, to marine environments such as on a naval vessel powered by gas turbines, a hybrid system or purely electric system, and to other transport applications including trains.

Claims

1. An electrical power system comprising: an electrical machine operable to output AC; a DC electrical network; a power electronics converter connected between the AC output of the electrical

machine and the DC electrical network and comprising a plurality of transistors and associated diodes connected in parallel with the transistors; and a controller configured to control switching of the transistors of the converter so that, during normal operation of the electrical power system, the converter rectifies the AC output of the electrical machine to supply the DC electrical network with DC electrical power, wherein: the controller is further configured, responsive to a determination to the effect there is a fault in the DC electrical network, to control a voltage source to inject a voltage to bias the diodes of the converter, and to control the switching of the transistors to control a level of current supplied to the faulted DC electrical network, and the voltage source comprises a resistor and a switch, and the switch (i) is closed to short the resistor during a normal operation of the DC electrical network so that current flows from the converter to the DC electrical network and (ii) is opened in response to the faulted DC electrical network so that current does not flow to the DC electrical network and is instead contained within the converter and the electrical machine operable to output the AC.

2. The electrical power system of claim 1, in which the voltage source further comprises one or more of the following connectable with the power electronics converter and the DC electrical network: a capacitor; and/or an inductor, wherein the switch in the voltage source is a semiconductor switching component.

3. The electrical power system of claim 1, wherein controlling the voltage source to inject the voltage to bias the diodes comprises actuating a semiconductor switching component, which is the switch, to connect the resistor to the DC electrical network.

4. The electrical power system of claim 1, in which the switch is a semiconductor switching component, and wherein controlling the voltage source to inject the voltage to bias the diodes comprises operating the semiconductor switching component in a linear operating region to produce the voltage to bias the diodes.

5. The electrical power system of claim 1, comprising a soft-start circuit, and wherein the soft-start circuit comprises the voltage source.

6. The electrical power system of claim 1, wherein controlling the switching of the transistors to control the level of current supplied from the electrical machine to the faulted DC electrical network comprises: switching the transistors of the converter to a crow-bar configuration in which current from the electrical machine does not flow to the DC network.

7. The electrical power system of claim 6, wherein controlling the switching of the transistors to control the level of current supplied from the electrical machine to the faulted DC electrical network further comprises: repeatedly switching the transistors between the crow-bar configuration and a rectifier configuration in which current from the electrical machine does flow to the DC network.

8. The electrical power system of claim 1, wherein controlling the switching of the transistors to control the level of current supplied from the electrical machine to the faulted DC electrical network comprises: modulating a pulse width modulation control parameter to control the level of current supplied from the electrical machine to the faulted DC electrical network.

9. The electrical power system of claim 1, further comprising a filter for smoothing a waveform of the current supplied to the faulted DC electrical network.

10. The electrical power system of claim 1, wherein the controller is further configured, responsive to the determination to the effect there is a fault in the DC electrical network, open one more circuit breakers or contactors in the DC electrical network.

11. The electrical power system of claim 1, in which: the transistors are IGBTs or MOSFETs and the diodes are separate from the transistors; or the transistors are MOSFETs and the diodes are body diodes of the MOSFETs.

12. The electrical power system of claim 1, wherein: the electrical machine has a plurality of phases; the power electronics converter comprises a phase leg for each one of the plurality of phases of the electrical machine, each phase leg comprising a first branch including a first transistor

and an associated parallel first diode connected between a phase connection of the electrical machine and a first DC output of the converter, and a second branch including a second transistor and an associated parallel second diode connected between the phase connection of the electrical machine and a second DC output of the converter, the plurality of transistors including the first and second transistors, and the associated diodes including the first and second diodes; and the first and second DC outputs of the converter are connected with the DC electrical network.

13. An aircraft power and propulsion system comprising: a gas turbine engine; and an electrical power system according to claim 1, wherein the electrical machine of the electrical power system is mechanically coupled with a spool of the gas turbine engine.

14. An aircraft comprising the power and propulsion system of claim 13.

15. A method of controlling an electrical power system comprising an electrical machine, a DC electrical network and a power electronics converter connected between the electrical machine and the DC electrical network, the method comprising: generating AC using the electrical machine and supplying the AC to an input of the converter; controlling switching of transistors of the converter to rectify the AC input to a DC output and supply the DC output to the DC electrical network; determining a condition to the effect that there is a fault in the DC electrical network; responsive to the determination, controlling a voltage source to inject a voltage to bias diodes of the converter; and after controlling the voltage source to inject a voltage to bias diodes of the converter, controlling switching of the transistors of the converter to control a level of current supplied to the DC electrical network, wherein controlling switching of the transistors of the converter to control the level of current supplied to the DC electrical network comprises modulating a pulse width modulation control parameter to control the level of current supplied from the electrical machine to the faulted DC electrical network, and wherein the voltage source comprises a resistor and a switch, the switch is a semiconductor switching component, and controlling the voltage source to inject the voltage to bias the diodes of the converter comprises operating the semiconductor switching component in a linear operating region to produce the voltage to bias the diodes.

16. The method of claim 15, wherein controlling switching of the transistors of the converter to control the level of current supplied to the DC electrical network comprises: switching the transistors of the converter to a crow-bar configuration in which current from the electrical machine does not flow to the DC network.

17. The method of claim 16, wherein controlling switching of the transistors of the converter to control the level of current supplied to the DC electrical network comprises: repeatedly switching the transistors between the crow-bar configuration and a rectifier configuration in which current from the electrical machine does flow to the DC network.

18. The method of claim 15, further comprising, after controlling switching of the transistors of the converter to control the level of current supplied to the DC electrical network: opening one more circuit breakers or contactors in the DC electrical network to isolate the fault in the DC electrical network.

19. A controller for an electrical power system configured to: during normal operation of the electrical power system, control switching of a plurality of transistors of a power electronics converter to rectify AC received by the converter from an electrical machine; and responsive to a determination to the effect there is a fault in a DC electrical network connected to a DC side of the converter, control a voltage source to inject a voltage to bias diodes of the converter, and control the switching of the transistors to control a level of current supplied to the faulted DC electrical network, wherein controlling the switching of the transistors to control the level of current supplied to the faulted DC electrical network comprises modulating a pulse width modulation control parameter to control the level of current supplied from the electrical machine to the faulted DC electrical network, and wherein the voltage source comprises a resistor and a switch, the switch is a semiconductor switching component, and controlling the voltage source to inject the voltage to bias

the diodes of the converter comprises operating the semiconductor switching component in a linear operating region to produce the voltage to bias the diodes.
